

AI and the Carbon Footprint

SMaiLE Project

Key Information

Target Group: 8 - 12 y.o.

Duration: 3 Weeks (4 lessons)

Key Learning Goals:

1. **Scientific Understanding:** Explain carbon footprints and AI energy consumption.
2. **Digital Awareness:** Evaluate environmental impact of digital habits.
3. **Critical Thinking:** Debate ethical AI use regarding the environment.
4. **Creative Solutions:** Propose AI solutions for reducing ecological footprints.

Learning Outcomes

Students will be able to:

KNOWLEDGE & UNDERSTANDING:

- Explain "CO₂ equivalents" and data center energy consumption.
- Understand the water and energy costs involved in AI operations.

SKILLS & ABILITIES:

- Compare energy costs of different digital actions (e.g., streaming vs. email).
- Work in teams to simulate energy use and solve challenges.
- Present sustainable AI-based ideas clearly.

ATTITUDES & VALUES:

- Demonstrate responsibility for digital habits.
- Value innovation as a tool for sustainability.

European Dimension / Erasmus+ Connection

- **Digital & Ecological Citizenship:** Building responsible digital citizens.
- **Transnational Challenges:** Comparing energy data across borders.
- **Innovation:** Encouraging creative technical solutions for green goals.



1. Resources and Tools

- **Research:** ChatGPT (Kids), WWF Footprint Calculator.
- **Creative Software:** Minecraft: Education Edition or SimCityEDU.
- **Materials:** "Data Packets" (cards/tokens), Art supplies.
- **Video:** "Can you see the cloud?" or similar about data centers.

Useful Links

- [Video: Hidden cost of digital life](#)
- [Video: Inside the Cloud](#)
- [Minecraft Education](#)
- [SimCity EDU article](#)

Classroom Support Materials

- Provide worksheet PDFs for the research phase and use Padlet or Google Forms for digital reflection and evaluation.

Activity Overview

Phase	Activity	Description
Intro	The Invisible Cost	Discussion: Brainstorming digital habits. Video on physical infrastructure of the internet.
Research	My Digital Footprint	Simulation: "Data Center Game" – role-playing servers to understand energy intensity. Calculating personal footprints.
Creative	AI Eco-Builder	Design: Using Minecraft or paper to build an "Eco-City" optimized by AI (smart transport, green energy).
Reflection	Action Plan	Evaluation: Self-reflection (Att 3.1) and class discussion on changing habits.

2. The Invisible Cost of the Internet

Goal: Make the invisible visible.

- **Concept:** The "Cloud" isn't in the sky; it's in a building.
- **Activity:** Watch a video tour of a data center.
- **Discuss:** How much water and electricity does it take to cool those machines?

3. My Digital Footprint

Goal: Experience energy consumption.

- **Role-Play:** Students act as servers. The teacher sends "requests" (cards) faster and faster.
- **Observation:** Servers heat up (get tired) and need "cooling" (fans/water).
- **Calculation:** Compare the footprint of sending an email vs. training an AI model.



4. AI Eco-Builder

Goal: Creative problem solving.

- **Task:** Design a sustainable city.
- **Tools:** Minecraft, SimCity, or Art supplies.
- **Key Elements:**
 - Efficient Transport (AI traffic lights).
 - Green Energy (Solar/Wind).
 - Waste Reduction (Recycling bots).
- **Present:** Share the city design and explain the AI features.

5. Reflection and Evaluation

Goal: Personal responsibility.

- **Self-Reflection:** Complete the reflection sheet (Att 3.1).
- **Discussion:** "What surprised you?" "What will you do differently?"
- **Action:** Pledge one eco-friendly digital habit (e.g., "I will delete old emails").